Pass-Through, Welfare, and Incidence under Product Returns

Draft notes

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Abstract

With product returns, the textbook local welfare formula $dCS = -D(p)\,dp$ overstates the consumer burden of a price change: only kept units matter, so $dCS = -D(p)(1 - r(p))\,dp$. Aggregates (D, r) are sufficient for this first-order measure. Pass-through is unchanged in form; incidence uses kept volume on the consumer side and producer surplus derived from profit maximization.

1 Setting

Consumers pay list price p at checkout, then learn a post-purchase match and may return the item at hassle cost κ_i. Producers are price-takers. Each firm j has production cost $C_j(q_j)$; returned units also cost h to handle. Write $r(p)$ for the aggregate return rate and $\mu(p) \equiv p - r(p)(p + h)$ for the effective margin per gross unit sold.

Equilibrium pairs list price p with gross quantity $Q = D(p)$ such that consumers optimize purchase and return decisions and producers optimize output given effective margin $\mu(p)$.

2 Consumer side

Each consumer type i has return hassle κ_i and draws match quality $u \sim G_i$ after purchase. Expected surplus from buying at list price p is

$$s_i(p) = \mathbb{E}_{u\sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

A type buys iff $s_i(p) \geq 0$; conditional on buying, it keeps the item iff $u \geq p - \kappa_i$, with keep probability $k_i(p) = \Pr(u \geq p - \kappa_i)$. Aggregating across types yields gross demand, kept volume, and the return rate:

$$D(p) = \int \mathbf{1}\{s_i(p) \geq 0\}\, dF(i), \qquad K(p) = \int \mathbf{1}\{s_i(p) \geq 0\}\, k_i(p)\, dF(i),$$
$$r(p) = 1 - K(p)/D(p).$$

Total consumer surplus is $CS(p) = \int \mathbf{1}\{s_i \geq 0\}\, s_i\, dF$. Differentiating along the price path, inframarginal buyers satisfy $s_i'(p) = -k_i(p)$, while marginal buyers have $s_i = 0$. By the envelope theorem,

$$CS'(p) = -K(p) = -D(p)\big(1 - r(p)\big), \tag{1}$$

so a small price change affects welfare only through kept units:

$$dCS = -D(p)\big(1 - r(p)\big)\, dp, \qquad \Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\, dp. \tag{2}$$

The weight on dp is kept volume K, not gross demand D.

3 Supply side

On the producer side, each firm j takes effective margin μ as given and chooses output to solve

$$\max_{q_j \geq 0} \; \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \;\; (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit:

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{3}$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve:

$$PS(\mu) = \int_0^{\mu} S(x)\, dx. \tag{4}$$

At equilibrium $\mu = \mu(p)$, $PS(\mu(p)) = \int_0^{\mu(p)} S(x)\, dx$.

4 Equilibrium

Combining the two sides, equilibrium requires gross demand to equal aggregated supply at effective margin $\mu(p)$:

$$D(p) = S\big(\mu(p)\big), \qquad Q \equiv D(p) = \sum_j q_j\big(\mu(p)\big). \tag{5}$$

Let j^* denote a marginal active firm ($q_{j^*} > 0$ and $\mu = C_{j^*}'(q_{j^*})$). Its per-unit production margin is zero in equilibrium:

$$m(p) \equiv \mu(p) - C_{j^*}'\big(q_{j^*}(\mu(p))\big) = 0. \tag{6}$$

Write $m_p \equiv dm/dp$ and $m_x \equiv \partial m/\partial x|_{Q,p}$. Differentiating the clearing condition and the zero-margin condition along the equilibrium path yields the pass-through objects used next.

5 Pass-through

Consider a small cost shock x, such as a uniform upward shift in every C_j' or an increase in handling cost h, and write $\rho_x \equiv dp/dx$. Holding (p, Q) fixed, a uniform production-cost shift lowers each firm's margin by one dollar on the marginal unit, so $m_x = -1$; a handling shock gives $m_x = -r(p)$. Differentiating market clearing and the zero-margin condition along the shocked equilibrium path then gives

$$\rho_x = \frac{S'(\mu)\, m_x}{S'(\mu)\, m_p - D'(p)}, \qquad \rho_h = r(p)\, \rho_C. \tag{7}$$

Here $S'(\mu) = \sum_j q_j'(\mu)$ is the aggregated slope of firm supply; no single aggregate $C(Q)$ is needed. In the homogeneous special case ($C_j = C$), $m_p = 1 - C''(Q)D'(p) - r'(p)(p + h) - r(p)$ and the formula above reduces to the standard competitive pass-through when $r \equiv 0$.

6 Incidence

Combining the consumer first-order formula $dCS = -K\, dp$ with the supply-side objects above, the welfare incidence of the shock is

$$\frac{dCS}{dx} = -K(p)\, \rho_x, \qquad \frac{dPS}{dx} = Q\big(m_p\, \rho_x + m_x\big). \tag{8}$$

The incidence ratio $I_x \equiv |dCS/dx|/|dPS/dx|$ is

$$I_C = I_h = \frac{(1 - r(p))\, S'(\mu)}{-D'(p)}. \tag{9}$$

Production and handling shocks bear the same consumer share because $\rho_h = r(p)\, \rho_C$ and the kept-volume factor $K = D(1 - r)$ applies to both.